

CHROMA

How a robot can learn to recognise things by touch — and why two thirds of this design currently works and one third does not.

Status	Research pilot. Sections 1–3 proved and confirmed; Section 4 failing its own tests.
Code	github.com/merolaagi/chroma-arch — MIT licence
Audience	No mathematics assumed.

The problem

Picture a robot arm in a wholesale grocery warehouse. Its job is to reach into a mixed pallet, feel around, and work out what it is touching before it grips. Cameras are little help: the bin is dark, items are shrink-wrapped and reflective, and half of them are buried under other items. So the arm carries eight small tactile pads, like fingertips, and must identify the item by touch alone.

Three things make this hard. Each one corresponds to a piece of the design.

Difficulty	Why it hurts
The robot moves	Every time a fingertip slides two centimetres, everything it feels changes. A ridge that was under the index pad is now under the middle pad.
Memory explodes	A cereal box feels completely different depending on which corner you touch and which way it is turned. One memory per viewpoint turns 1,800 products into hundreds of thousands of stored patterns.
The warehouse changes	In summer the pallets hold soft produce and thin bags; in winter, boxed dry goods and cans. The robot's sense of how things usually feel has to shift — slowly, and not because of one odd item.

Three ideas

1. Geometry is arithmetic, not something to learn

Think about reading braille. If you know your finger moved exactly one character to the right, you know precisely how the dot pattern should shift. You do not guess. You do not need training examples. It is bookkeeping.

CHROMA builds that bookkeeping in as a hard rule. When the gripper moves, the robot's internal picture of what it is touching is shifted by exactly the right amount — computed, never learned. The learned part covers only what geometry cannot explain: the item was squishier than expected, the wrap slipped, there was a seam.

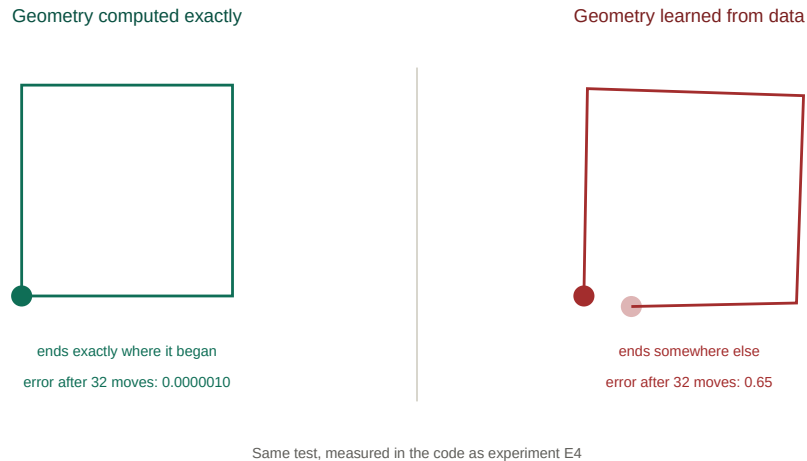


Figure 1. The test: trace a square and return to the start. Exact geometry returns to within rounding error. Learned geometry ends up somewhere else.

2. Store the object, not the photographs

Rather than recording what a noodle carton feels like from five hundred angles, CHROMA stores one description in the carton's own coordinates: corner here, seam there, ridged panel over there. When a fingertip makes contact at some arbitrary angle, the geometry bookkeeping translates the sensation back into carton coordinates before comparing.

It is the difference between a floor plan of a building and five hundred photographs taken from different doorways. With a floor plan you only have to walk the building once.

3. A slow dial that gates everything

The robot carries a small internal state that changes over weeks, not milliseconds. Think of a chef who has decided that tonight is Italian, versus one who re-evaluates their entire technique after every single ingredient.

This state does not identify items. It decides which of the robot's skills are turned up, which are turned down, and — the interesting part — which skills are still allowed to change. A grip strategy that has worked reliably for months gets locked. One that has started failing gets unlocked and becomes trainable again. The idea is borrowed from how a stem cell commits to becoming a specific cell type, and can un-commit under injury.

How the three fit together

Three loops run at three speeds. One quantity connects them all: the **prediction error** — the gap between what the robot expected to feel and what it actually felt. That single number trains the fast loop, weights the middle loop, and steers the slow one.

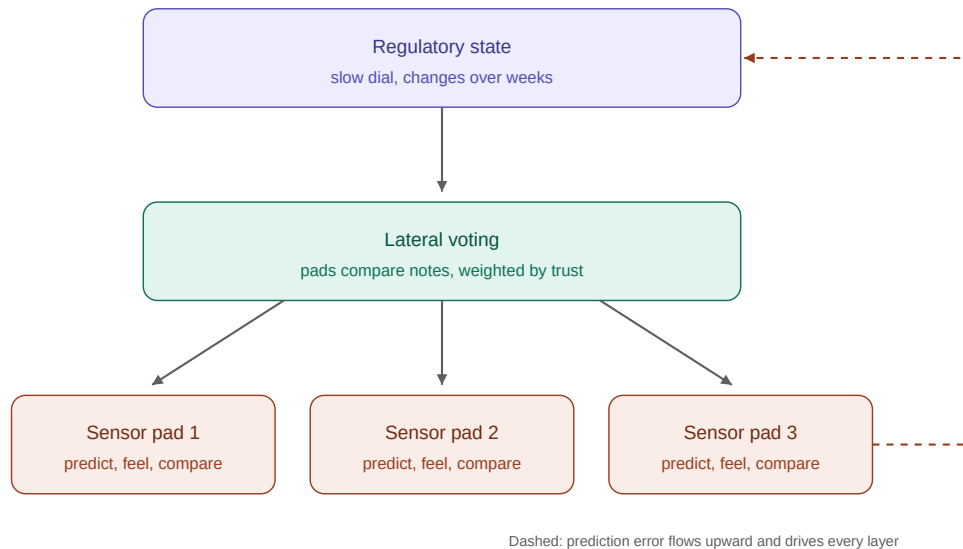


Figure 2. The fast loop predicts and feels; the middle loop pools opinions across pads; the slow loop sets the operating mode.

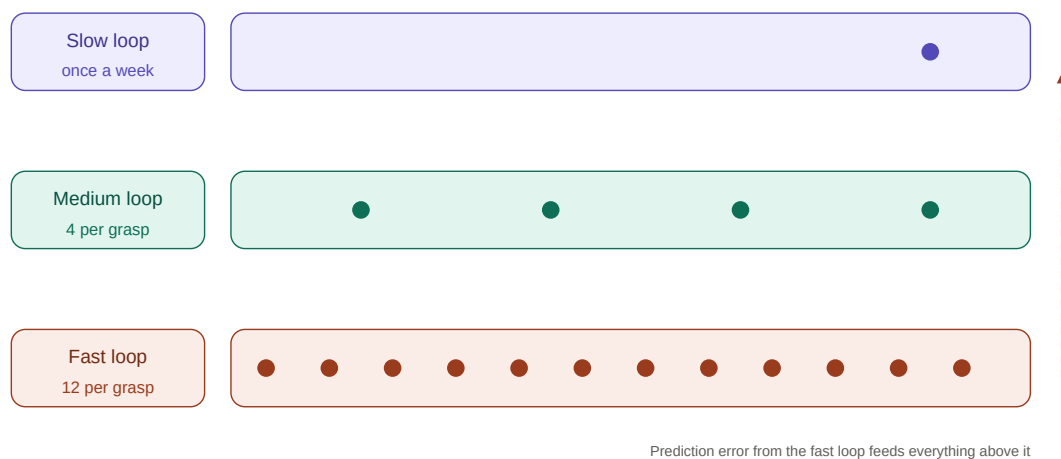


Figure 3. The same three loops seen as timing. One grasp is twelve finger movements; the slow dial barely moves in a week.

One grasp, step by step

The gripper descends into a mixed pallet. Target: a twelve-pack of instant noodle cups, shrink-wrapped, partly under a bag of rice.

Step	What happens
0. Contact	Eight pads touch the item at eight spots. Each pad knows exactly where it sits relative to the others — that is fixed by the hardware, so it is given, not learned.
1. The guess	Pad 3 feels something ridged. Before moving, the robot commits to a prediction: slide two millimetres and this ridge should shift by exactly this much, with the texture unchanged. The geometric half is computed exactly; only the texture half is a learned guess.
2. Reality check	The hand moves and the pad reports what it actually feels. The gap is the prediction error. Note that the robot predicts at the level of “ridged cardboard edge”, not raw pressure readings — the plot of a film, not every pixel.
3–6. Evidence	Each pad translates its sensation back into object coordinates and asks the catalogue for a match. Pad 3 says “noodle carton, rotated about thirty degrees, left face”. Pad 7 is torn between two products.
7. Compare notes	Because the hand geometry is known, a confident pad's guess can be translated into any other pad's frame. They pool evidence, weighted by how well each has been predicting lately. A pad with high recent error is discounted automatically. Recognition usually settles here.
8–12. Grip	Remaining moves confirm the hypothesis and locate the seam for a safe grip.
Once a week	The slow dial has been quietly integrating average error and disagreement across thousands of grasps. As the seasonal mix shifts, skills that keep working stay locked; skills that start failing unlock and retrain.

That last row addresses a real and expensive problem. Retrain a normal robot on new items and it often gets *worse* at the old ones — catastrophic forgetting. The locking mechanism here comes with a mathematical guarantee: as long as a skill keeps performing, how far it can drift is capped by a fixed number that does not grow with time, no matter how many months pass.

What breaks if you remove each piece

Remove	What happens
Exact geometry	The hand loses track of where it is after about ten moves. Errors compound and long exploration becomes useless.
Object-coordinate memory	Every item must be felt from every angle. 1,800 products becomes an impossible amount of training.
Multiple voting pads	One misleading contact — a fingertip on a label instead of cardboard — throws the whole identification.
The slow dial	The robot re-tunes its entire strategy every time it touches something unusual. It never settles.
Plasticity locking	Teaching it this season's items degrades last season's. Every changeover costs accuracy on things it already knew.

Where it actually stands

Ideas 1 and 2 are proved, both on paper and in the running code. They are not especially new; the contribution is welding them together. Idea 3 is the invented part, and the honest result so far is that it mostly does not work as claimed.

The decisive test compares three versions: the full system; a version with the “settles into distinct modes” property removed but the slowness kept; and a version with neither. The middle version performs identically to the full one.

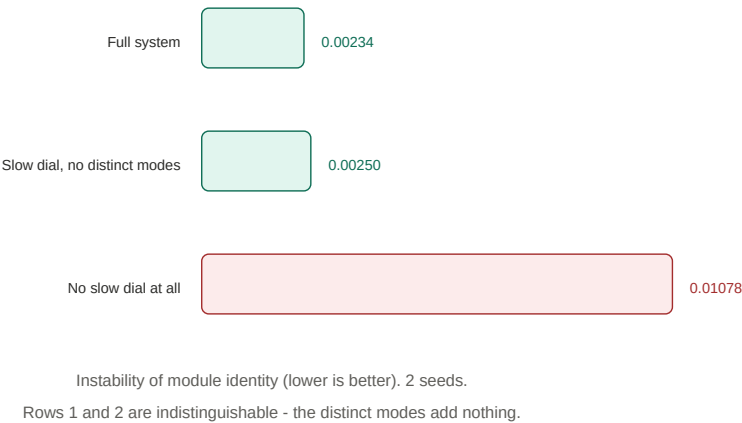


Figure 4. Instability of module identity across the three versions. Lower is better. The top two bars are the same height.

The benefit therefore comes from the dial simply being **slow** — not from it having distinct settings, which was the actual claim. A plain running average would deliver the same thing with none of the biology-inspired machinery. A second test, asking whether the eight pads specialise into distinct roles over time the way stem cells differentiate, also came back negative: they stayed nearly identical.

Both results come from a run at one fiftieth of the intended size, so they may yet flip. But as things stand, the most elaborate part of the design is the part failing its own test, and the plain parts are the ones carrying the system.

Scoreboard

Claim	Verdict	Evidence
Exact geometric transport	Supported	Error 0.0000010 vs 0.65 for the learned alternative
Viewpoint-free memory	Supported	Proved, and confirmed by test
Bounded forgetting	Supported in isolation	Measured drift 19.53 against a predicted ceiling of 20.0
Distinct modes help	Not supported	Flat-landscape version ties the full system
Slow integration helps	Weakly supported	4.6x better than no dial; the threshold set in advance was 5x
Pads specialise over time	Not supported	0.214 against a pre-set threshold of 0.7

Reproducing this

Everything above is generated by the repository, and every claim on paper is an executable test. The full comparison is fifteen runs — five random seeds across three versions — which is about one overnight session on a desktop machine.

```
git clone https://github.com/merolaagi/chroma-arch
cd chroma-arch && pip install torch
PYTHONPATH=. python tests/test_propositions.py
python experiments/e1_to_e4.py e4
python run_e1_chunk.py chroma 20000 0
python aggregate.py
```

If the flat-landscape version still ties the full system at full scale, the right response is to delete the elaborate machinery and keep the timescale. That would be a smaller claim than the one the design started with, and a true one.